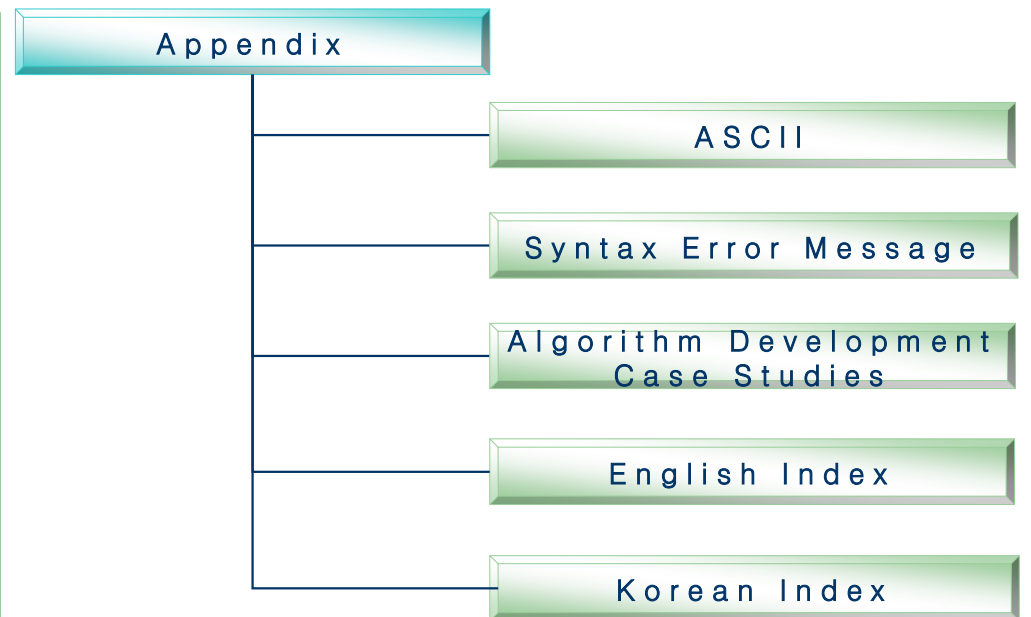


Appendix

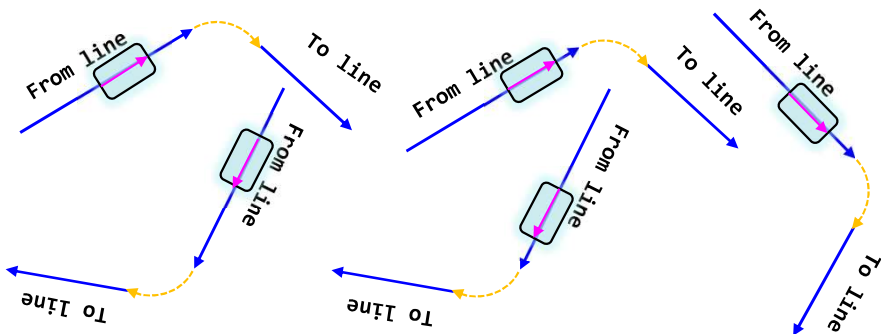


1 An algorithm is introduced that automatically calculates the turning curvature and turning direction when straight-line segments are given for applications such as highway alignment or robot navigation paths. The algorithm presented in this book is called AA-1 (Angel Algorithm-1) and is made publicly available for broad use. In addition, the geometric formulas used in this algorithm can serve as fundamental formulas for solving a wide variety of geometric problems. Suggestions for improvement are welcome.

As shown in the following figure, when a moving object is given the coordinates of a From Line ($X1, Y1, X2, Y2$) and a To Line ($X1, Y1, X2, Y2$), the following information can be calculated using the geometric properties of the two lines:

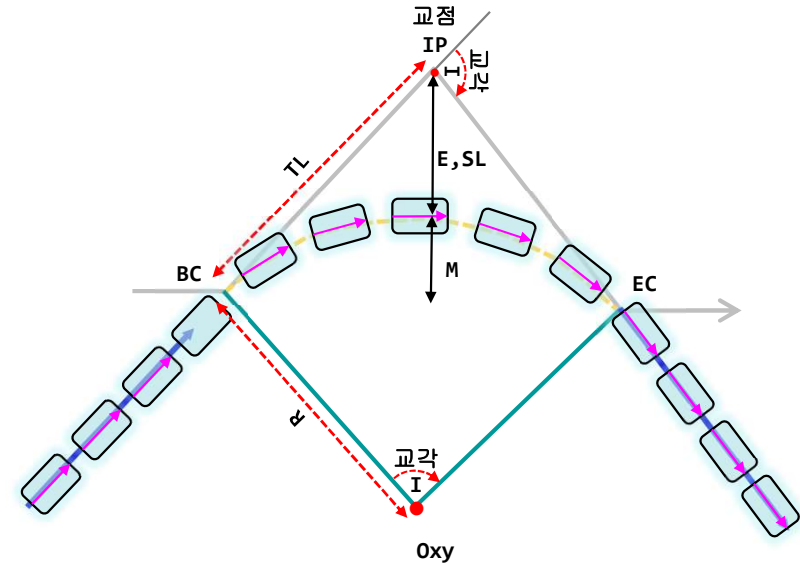
- Whether the two lines intersect and the coordinates of the Intersection Point (IP)
- Geometric properties: Tangent Length (TL), Center Point, and Radius of Curvature (R)
- Turning direction for the curved path

The main difficulty in developing this algorithm was establishing a general rule that could handle all possible combinations of line directions, as illustrated below.



DESCRIPTION

Understanding the Basic Formulas



$$\text{접선길이}(TL) = R \tan \frac{I}{2}$$

$$\text{곡선길이}(CL) = R \cdot I \cdot \frac{\pi}{180^\circ} \cdot \frac{I}{360^\circ} = \frac{CL}{2\pi R}$$

$$\text{외할}(E \text{ 또는 } SL) = R \left(\sec \frac{I}{2} - 1 \right)$$

$$\text{중양종거}(M) = R \left(1 - \cos \frac{I}{2} \right)$$

$$\text{현의 길이}(C) = 2 \left(R \sin \frac{I}{2} \right)$$

$$\text{편각}(\delta) = \frac{I}{2R} \times \frac{180^\circ}{\pi}$$

$$\text{곡선의시점}(BC) = IP - TL = IP - \left(R \tan \frac{I}{2} \right)$$

$$\text{곡선의종점}(EC) = BC + CL = IP - TL + CL = IP - \left(R \tan \frac{I}{2} \right) + \left(R \cdot I \cdot \frac{\pi}{180^\circ} \right)$$

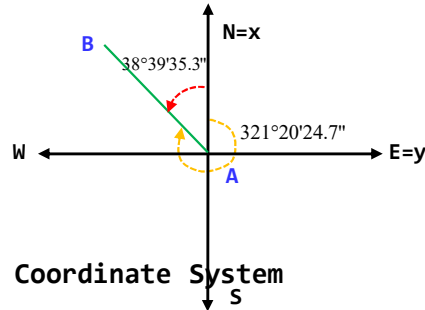
1

AB Direction Calculation : Geomatics Coordinate System

$$= \tan^{-1} \left(\frac{y_B - y_A}{x_B - x_A} \right) = \tan^{-1} \left(\frac{7-15}{20-10} \right)$$

$$= N38^\circ 39' 35.3'' W (4\text{상한})$$

Coordinate	B(End)	A(Start)
X	10	20
y	15	7



AB Azimuth Calculation : Geomatics Coordinate System

$$= 360^\circ - N38^\circ 39' 35.3'' W (4\text{상한})$$

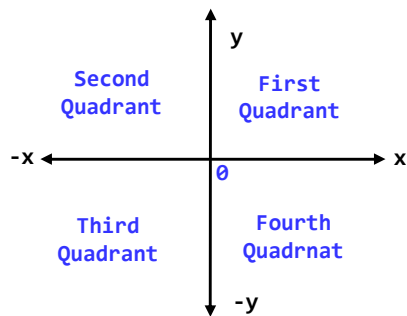
$$= 321^\circ 20' 24.7''$$

Quadrant	Azimuth(α)	Direction
First	$0^\circ \sim 90^\circ$	N α E
Second	$90^\circ \sim 180^\circ$	S $(180^\circ - \alpha)$ E
Third	$180^\circ \sim 270^\circ$	S $(\alpha - 180^\circ)$ W
Fourth	$270^\circ \sim 360^\circ$	N $(360^\circ - \alpha)$ W

Back Azimuth Calculation

Add 180° to the azimuth. If the result is 360° or greater, subtract 360° ; otherwise, use the resulting angle as is.

Note: Mathematical Coordinate System



$$= \tan^{-1} \left(\frac{x_B - x_A}{y_B - y_A} \right)$$

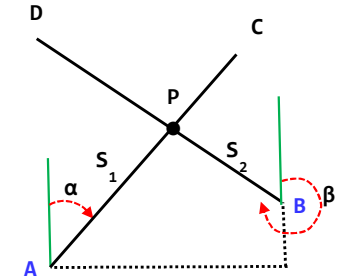
Understanding Line Intersection Calculation

Point A $P_x = A_x + (S_1 \cdot \cos \alpha)$

$$P_y = A_y + (S_1 \cdot \sin \alpha)$$

Point B $P_x = B_x + (S_2 \cdot \cos \beta)$

$$P_y = B_y + (S_2 \cdot \sin \beta)$$



Point P can be calculated using the following equation

$$A_x + S_1 \cdot \cos \alpha = B_x + S_2 \cdot \cos \beta$$

$$A_y + S_1 \cdot \sin \alpha = B_y + S_2 \cdot \sin \beta$$

$$S_1 \cdot \cos \alpha - S_2 \cdot \cos \beta = B_x - A_x = \Delta X$$

$$S_1 \cdot \sin \alpha - S_2 \cdot \sin \beta = B_y - A_y = \Delta Y$$

$\sin \beta$ 또는 $\cos \beta$ 를 곱하여 연립방정식으로

$$S_1 \cdot \cos \alpha \sin \beta - S_2 \cdot \cos \beta \sin \beta = \Delta X \cdot \sin \beta$$

$$S_1 \cdot \sin \alpha \cos \beta - S_2 \cdot \cos \beta \sin \beta = \Delta Y \cdot \cos \beta$$

$$S_1 (\cos \alpha \cdot \sin \beta - \sin \alpha \cos \beta) = \Delta X \cdot \sin \beta - \Delta Y \cdot \cos \beta$$

$$S_1 = \frac{\Delta X \cdot \sin \beta - \Delta Y \cdot \cos \beta}{\cos \alpha \cdot \sin \beta - \sin \alpha \cdot \cos \beta}$$

삼각함수 덧셈공식

$$\sin(A - B) = \sin A \cdot \cos B - \cos A \cdot \sin B \quad \text{용}$$

$$S_1 = \frac{\Delta y \cdot \cos \beta - \Delta x \cdot \sin \beta}{\sin(\alpha - \beta)}$$

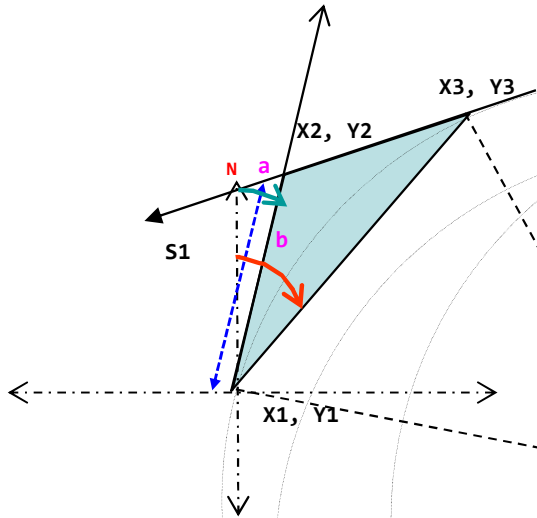
$$S_2 = \frac{\Delta y \cdot \cos \alpha - \Delta x \cdot \sin \alpha}{\sin(\alpha - \beta)}$$

2 [Implementation Process : AA-1_LR_Turn_2026_08_17.c]**0. Start**

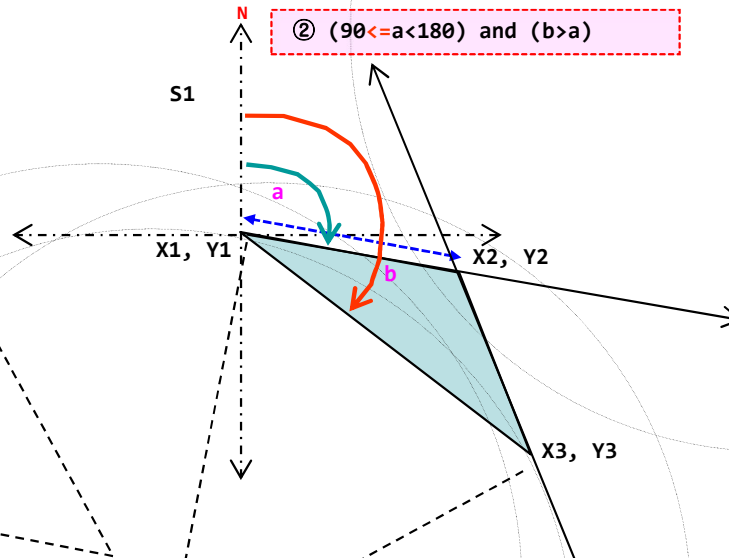
- 1. From Line / To Line Coordinate Input**
- 2. Check for Intersection of Two Lines**
- 3. Calculation of Intersection Point(IP)**
- 4. Distance Calculation from the Intersection Point**
- 5. Determine the Shorter Tangent Length(TL)**
- 6. Calculate the Azimuths (a, b) of the Two Lines**
- 7. Intersection Angle(I) Calculation**
- 8. Radius of Curvature(R) Calculation**
- 9. Determine the Turning Direction(CW/CCW)
Using a and b**
- 10. Direction Angle Calculation**
- 11. Center of Curvature(Oxy) Calculation**
- 12. Divide the Turning Angle into Equal Increments**
- 13. Generate ARC Coordinate**
- 14. End**

3 Draw a triangle using the intersection and starting points based on the shorter distance(CW)

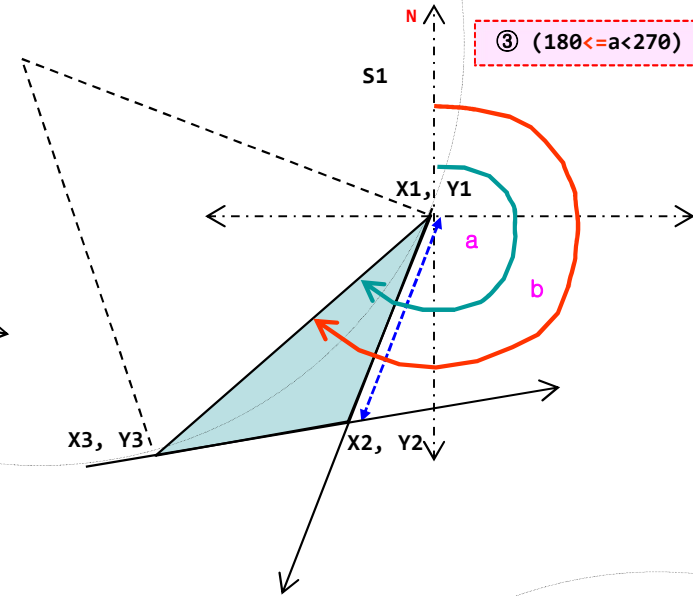
① ($0 \leq a < 90$) and ($b > a$)



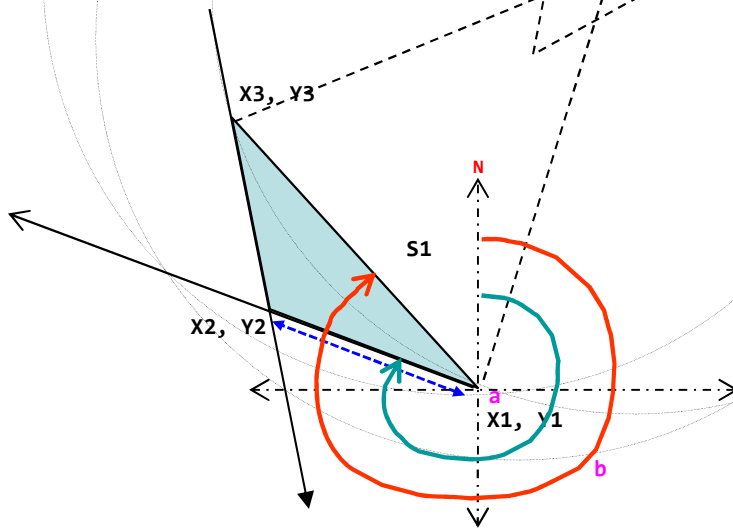
② ($90 \leq a < 180$) and ($b > a$)



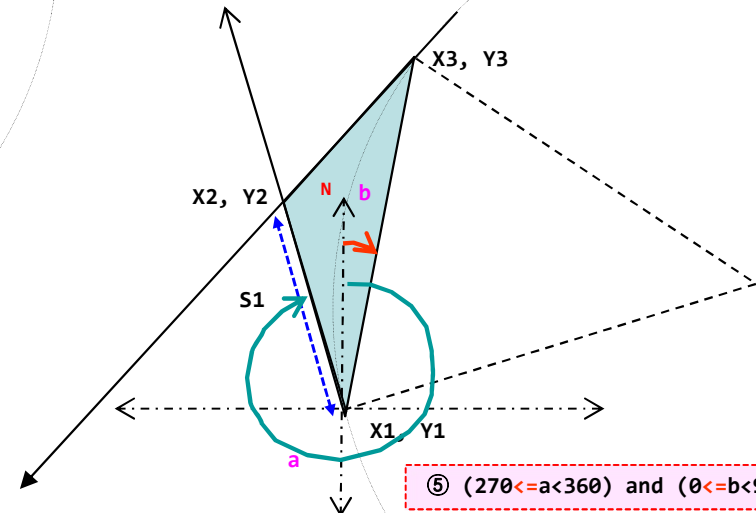
③ ($180 \leq a < 270$) and ($b > a$)



④ ($270 \leq a < 360$) and ($270 \leq b < 360$) and ($b > a$)



⑤ ($270 \leq a < 360$) and ($0 \leq b < 90$) and ($a > b$)



4 Draw a triangle using the intersection and starting points based on the shorter distance(CCW)

